Exp 5: Remote File Access via FTP/WinSCP & Git Init/Push to GitHub

```
# 1. On Ubuntu VM: Navigate to folder and create test file
mkdir -p ~/exp5 && cd ~/exp5
echo "Hello World" > file1.txt

# 2. Configure Git Identity
git config --global user.name "codedbycliff"
git config --global user.email "rodriguescliff7@gmail.com"

# 3. Initialize Git repo and commit
git init
git status
git add file1.txt
git commit -m "initial commit for exp5"
git branch -M main

# 4. Link to GitHub repository and push
git remote add origin https://github.com/codedbycliff/test.git
git push -u origin main --force
# (Enter GitHub username and Personal Access Token when prompted)

# Note: For Remote File Access, connect FileZilla or WinSCP using:
# Host: 127.0.0.1 (or VM IP), Port: 2222 (or 22), User: vboxuser
```

Exp 6: Remote Access & Git Sync (Host & VM + WinSCP/FileZilla Validation)

```
# 1. On VM: Clone the repository
cd ~
git clone https://github.com/codedbycliff/test.git
cd test

# 2. Check status and push updates from VM / Host
git status
git add .
git commit -m "Sync update"
git push origin main

# 3. Pull latest changes to synchronize
git pull origin main

# 4. WinSCP / FileZilla Validation:
# Open WinSCP/FileZilla -> Connect to 127.0.0.1:2222 -> Navigate to
/home/vboxuser/exp5/
# Verify file1.txt and test directory are visible and synchronized.
```

Exp 7: Git Multi-User Collaboration & Synchronization (Access Server)

```
# 1. Collaborator (Friend) clones repo and makes changes (on Host or second terminal):
git clone https://github.com/codedbycliff/test.git
cd test
echo "My First commit for the lab collab project." > readme.md
git add .
git commit -m "Add initial README file"
git push origin main
# (Authenticate with collaborator credentials)

# 2. Owner pulls changes on VM:
cd ~/exp5
git pull origin main
cat readme.md
```

Exp 8: Static Web Deployment on Vercel & Remote MySQL Server Config

```
# PART A: HTML Page Creation & GitHub/Vercel Deployment
mkdir -p ~/portfolio && cd ~/portfolio

cat << 'EOF' > index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Cloud Portfolio</title>
  <style>
    body { font-family: sans-serif; text-align: center; margin-top: 50px; }
    h1 { color: #2c3e50; }
  </style>
</head>
<body>
  <h1>Hello, cloud!</h1>
  <p>This page is deployed from a GitHub repository using Vercel.</p>
  <p>Student Name: Salil Pai Raikar</p>
</body>
</html>
EOF

# Push to GitHub
git init
git add .
git commit -m "Add initial portfolio page"
git branch -M main
git remote add origin https://github.com/codedbycliff/portfolio.git
git push -u origin main
# Import repo on https://vercel.com and deploy to get live URL.

# PART B: Install MySQL & Configure Remote Database
sudo apt install mysql-server -y
sudo mysql -u root

-- Inside MySQL Shell:
CREATE DATABASE cc_db;
USE cc_db;
CREATE TABLE users (
  id INT PRIMARY KEY,
  name VARCHAR(30),
  email VARCHAR(50),
  contact VARCHAR(12),
  address VARCHAR(200)
);
INSERT INTO users (id, name, email, contact, address) VALUES (69, 'Vinay',
'vinaygupta0758@gmail.com', '1234567890', 'Panjim');
SELECT * FROM users;
```

```
-- Enable remote user and grant privileges:
CREATE USER 'remoteuser'@'%' IDENTIFIED BY 'vboxuser';
GRANT ALL PRIVILEGES ON *.* TO 'remoteuser'@'%' WITH GRANT OPTION;
FLUSH PRIVILEGES;
EXIT;

-- Edit MySQL bind-address to allow remote connections:
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
# Change: bind-address = 127.0.0.1  -->  bind-address = 0.0.0.0
sudo systemctl restart mysql
```

Exp 9: Split Deployment - Vercel Form Frontend + Ngrok Tunnel + MySQL Backend

```
# 1. Install Apache, PHP, MySQL stack
sudo apt install apache2 php libapache2-mod-php mysql-server php-mysql -y

# 2. Create Database and Table in MySQL
sudo mysql -u root

-- Inside MySQL:
CREATE DATABASE web_project;
USE web_project;
CREATE TABLE contacts (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(50),
  email VARCHAR(100),
  message TEXT
);
EXIT;

# 3. Create Backend PHP Script (/var/www/html/submit.php)
sudo tee /var/www/html/submit.php << 'EOF'
<?php
$conn = new mysqli("localhost", "root", "", "web_project");
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
$name = $_POST['name'];
$email = $_POST['email'];
$message = $_POST['message'];

$sql = "INSERT INTO contacts (name, email, message) VALUES ('name', 'email', '$message')";
if (conn -> query(sql) === TRUE) {
  echo "Data saved successfully!";
} else {
  echo "Error: " . $sql . "<br>" . $conn->error;
}
$conn->close();
?>
EOF

# 4. Install and Start Ngrok Tunnel (to expose local Apache to Vercel frontend)
mkdir -p ~/ngrok-setup && cd ~/ngrok-setup
wget https://bin.equinox.io/c/bNyj1mQVY4c/ngrok-v3-stable-linux-amd64.tgz
tar -xvzf ngrok-v3-stable-linux-amd64.tgz
sudo mv ngrok /usr/local/bin/

# Authenticate ngrok and expose Port 80:
# (Replace with your own authtoken if needed)
ngrok config add-authtoken 3K3NP7STN3MNE6L35r6x130b2eP_5uAqExGEy8Uq8cyn9m513
```

```
ngrok http 80
# Copy the Forwarding HTTPS URL: e.g., https://your-subdomain.ngrok-free.dev

# 5. Create Frontend HTML Form (Deploy this repo to Vercel)
# Replace the action URL with your ngrok forwarding URL:
cat << 'EOF' > index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Contact Form</title>
</head>
<body>
  <h1>Contact Us</h1>
  <form action="https://your-subdomain.ngrok-free.dev/submit.php" method="POST">
    <input type="text" name="name" placeholder="Enter your name" required><br><br>
    <input type="email" name="email" placeholder="Enter your email" required><br><br>
    <textarea name="message" placeholder="Enter your message"></textarea><br><br>
    <button type="submit">Submit</button>
  </form>
</body>
</html>
EOF

# 6. Verify Form Submission in MySQL Backend:
sudo mysql -u root -e "USE web_project; SELECT * FROM contacts;"
```